

- 7 (a)** The resistance of the sample increases linearly from $-200\text{ }^{\circ}\text{C}$ to $15\text{ }^{\circ}\text{C}$, decreases at an increasing rate from $15\text{ }^{\circ}\text{C}$ to $50\text{ }^{\circ}\text{C}$, decreases linearly from $50\text{ }^{\circ}\text{C}$ to $100\text{ }^{\circ}\text{C}$, and decreases at an increasing rate from $50\text{ }^{\circ}\text{C}$ to $100\text{ }^{\circ}\text{C}$.

(b) To show $R \propto \frac{1}{\theta}$, i.e. $R\theta = \text{constant}$.

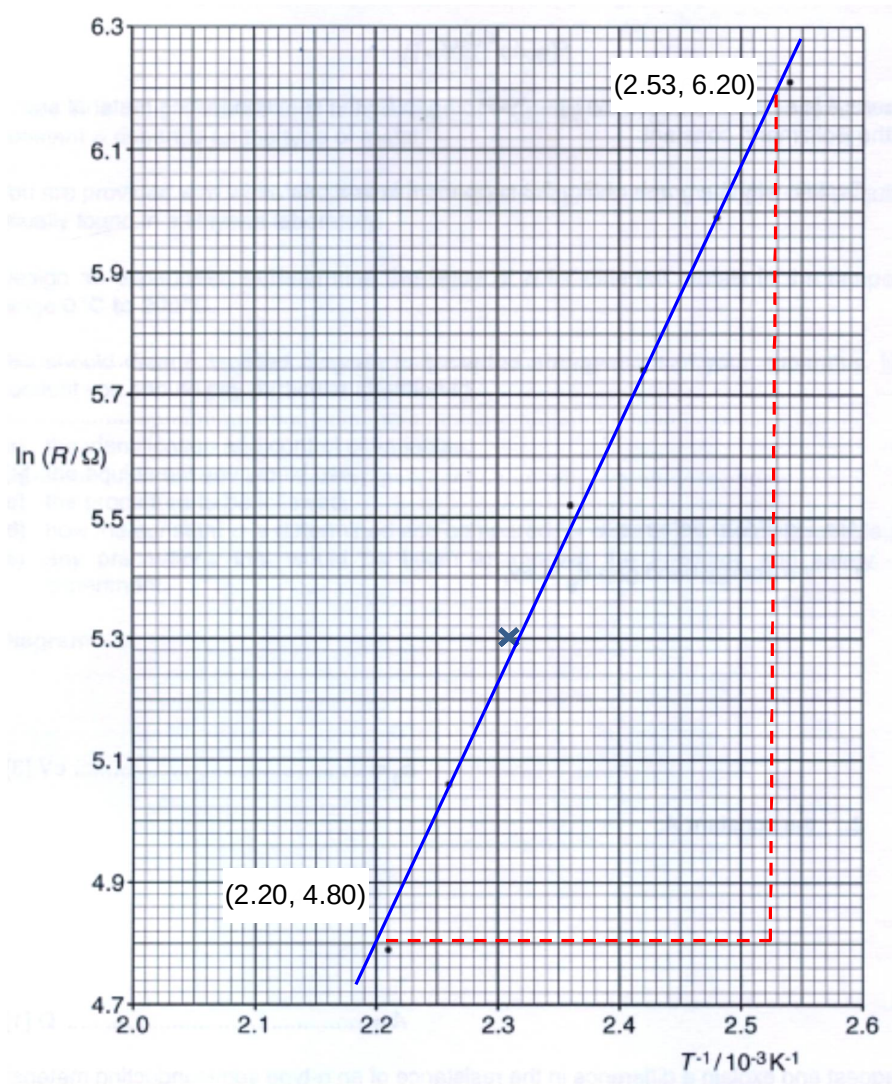
θ	R	$R\theta$
80	1200	96000
100	800	80000
120	500	60000

Since $R\theta$ is not a constant from 50 °C to 150 °C, the proposal is not correct.

(c) (i)

R / Ω	$\theta / ^\circ\text{C}$	$T^{-1} / 10^{-3} \text{ K}^{-1}$	$\ln(R / \Omega)$
200	160	2.31	5.30

(ii)



(d) (i) $R = Ae^{\frac{E_g}{2kT}}$

Taking natural logarithm on both sides of equation,

$$\ln R = \ln A + \frac{E_g}{2kT}$$

$$\ln R = \frac{E_g}{2k}(T^{-1}) + \ln A$$

Hence, the equation has the form of a straight line graph $y = mx + c$, as is the graph of $\ln R$ vs T^{-1} which is a straight line with gradient $\frac{E_g}{2k}$ and vertical intercept $\ln A$. Hence, the graph of Fig. 7.3 supports this proposal.

- (ii) 1. Using gradient coordinates $(2.20 \times 10^{-3}, 4.80)$ and $(2.53 \times 10^{-3}, 6.20)$,

$$\text{Gradient} = \frac{6.20 - 4.80}{(2.53 - 2.20) \times 10^{-3}}$$

$$\frac{E_g}{2k} = \frac{6.20 - 4.80}{(2.53 - 2.20) \times 10^{-3}}$$

$$E_g = \frac{6.20 - 4.80}{(2.53 - 2.20) \times 10^{-3}} \times \frac{2k}{e} = 0.732 \text{ eV}$$

2. Using $(2.20 \times 10^{-3}, 4.80)$,

$$\begin{aligned} \ln A &= \ln R - \frac{E_g}{2k}(T^{-1}) \\ &= \ln R - (\text{gradient})(T^{-1}) \\ &= 4.80 - (4242)(2.20 \times 10^{-3}) \\ &\approx 0.0108 \Omega \end{aligned}$$

- (e) An n-type semiconductor has a donor energy level near the bottom of the conduction band. Hence, it is easier for electrons to be thermally excited from this donor energy level to the conduction band. This results in a much larger charge carrier density (free electrons) and lower resistance in an n-type semiconductor compared to an intrinsic semiconductor.
- (f) A metal has no band gap because the conduction band is partially filled. Hence, temperature has negligible effect on charge carrier density in a metal. On the other hand, an increase in temperature causes an increase in the ionic lattice vibrations, which results in increased collisions with the drifting electrons thus impeding their motion and increasing the resistance of the metal.

8. Diagram (equipment to be used is shown in the diagram)